
Shibboleth: Probe-and-Replay Model-Response Watermarking for Diffusion Language Models

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Abstract

1 Text watermarks usually hide a secret in token choices and later test those tokens
2 without rerunning the generator. We ask whether a masked diffusion language
3 model (DLM) can instead help verify its own output. Hiding selected parts of the
4 context changes the probability of an emitted token in a reproducible, checkpoint-
5 specific direction. Shibboleth uses that model response as the watermark carrier.
6 The method first finds positions where plausible candidates exhibit both response
7 directions. Only after a position is fixed does the key choose which direction to
8 favour. The generator then resamples from its current distribution without changing
9 the decoding schedule. To verify a finished document, Shibboleth reconstructs
10 the same masked contexts and counts how often the responses agree with the
11 key. Position selection cannot observe the requested direction, so text produced
12 independently of the key gives an exact binomial count.
13 At 1% FPR, Shibboleth reaches 0.80 TPR on LLaDA-8B-Instruct and 0.90 on
14 Dream-7B-Instruct. The strongest token-local baseline reaches 0.93 on LLaDA
15 but 0.57 or less on Dream when its LLaDA setting is transferred unchanged. At
16 Shibboleth’s main setting, the PPL ratio is $0.74\times$ on LLaDA and $1.09\times$ on Dream,
17 and changes across the measured task scores are at most 0.06. This checkpoint
18 binding has a cost: verification needs the same model and 144 full-sequence calls
19 per 512-token document, and dispersed edits reduce detection power.

20 1 Introduction

21 When language-model text resembles human writing, its origin can be difficult to establish in settings
22 such as disinformation analysis, phishing investigation, and academic-integrity review [28]. A
23 post-hoc detector learns recurring output patterns; a watermark instead changes generation under
24 a secret key and lets the key holder test a later document [21, 33]. We focus on watermarking
25 because it can offer a stated finite-sample false-positive rate, rather than relying on a classifier whose
26 behaviour may change with the decoder [10, 19, 23, 44].

27 Nearly all language-model watermarks place the secret in token identities: the key biases a vocabulary
28 subset or couples randomness to each draw. Their detector therefore needs only the text and key
29 (Figure 1a) [21, 24]. This design is inexpensive, but it must coexist with the decoder. For masked
30 DLMs, that decoder may resolve many positions in parallel and choose their order from confidence,
31 so forcing a resolved prefix or steering the order can change the system being watermarked [17].

32 This tension suggests a different question: can the originating model, rather than the token alone,
33 carry the signal? A masked DLM can answer a reproducible question about its own output. Take a
34 completed sentence, hide one subset of its earlier context, and record the probability of an emitted
35 token; then hide a second subset and record it again. Some candidates become more likely and others
36 less likely. The sign of that change is a *model response*. Because the verifier can recreate both hidings

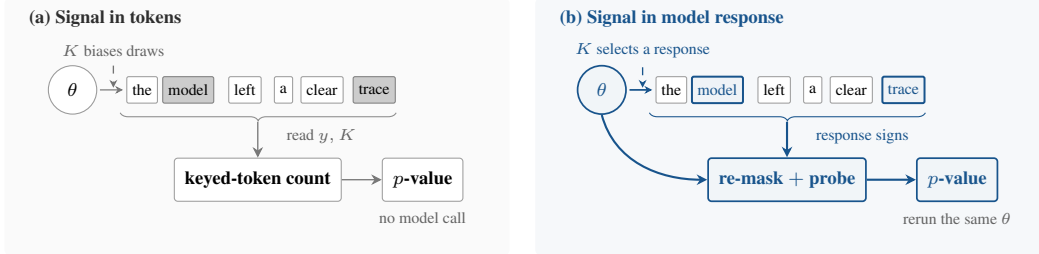


Figure 1: **Two places to encode a watermark.** Token-local methods alter sampled identities and verify from y, K alone. Shibboleth instead encodes which context response the checkpoint favours, so verification reruns the same θ .

from the finished text, this response can carry a keyed signal even when the token identity alone does not (Figure 1b).

Two obstacles separate this observation from a valid detector. First, verification must recreate the generation-time state even though the full continuation is now visible. Second, the method may seek positions where both response directions are plausible, but it must finish that search before the key reveals which direction counts as a match. Reversing this order would bias the selected positions toward the requested answer.

Shibboleth addresses both obstacles. For each active block, it re-masks that block and its suffix, compares pairs of hidden-prefix probes, and selects positions that retain probability mass on both response directions. The key then supplies the requested direction. At a selected position, Shibboleth draws up to R candidates from the decoder’s current distribution and commits the first match; all other positions follow the reference decoder. Verification reconstructs the probes, repeats the selection rule, and evaluates an exact binomial tail. The block partition, confidence order, and checkpoint probabilities remain those of the reference decoder.

Our contributions are:

- We introduce model-response watermarking: the signal is a checkpoint’s reproducible reaction to hidden context rather than a keyed pattern in token identities.
- We give a replayable construction whose carrier positions are selected before the keyed orientation is consulted, yielding finite-sample false-positive control while preserving the reference decoding schedule (Sections 3.2–3.5).
- We evaluate two masked DLMs. At 1% FPR, Shibboleth reaches 0.80 TPR on LLaDA and 0.90 on Dream. Verification requires 144 full-sequence model calls, and detection power decreases under dispersed edits (Section 4).

2 Related Work

Masked diffusion language models. Masked DLMs generate text by repeatedly predicting unresolved positions and committing a subset of them. This extends discrete diffusion to text [3, 16, 29, 32, 37, 38] and now supports instruction-following models including LLaDA and Dream [25, 31, 42, 45]. Which positions commit, and in what order, affects both quality and speed [2, 6, 20, 26, 39, 40]. We therefore treat the deployed schedule as a constraint: Shibboleth observes the model’s masked-token predictions but does not choose a new commit order.

Watermarks encoded in tokens. Autoregressive watermarks usually bias a keyed vocabulary subset [21] or couple each draw to keyed randomness [1, 13, 24]. Their detectors are cheap because a key–token calculation needs no model call, and later work characterises when those detectors are reliable or undetectable [10, 22, 44]. DLM methods adapt this idea by removing prefix dependence, averaging a red–green bias over unresolved context, sampling with keyed randomness, or encoding information in the denoising order [4, 9, 14, 17, 34, 41]. Our closest baseline, dgMARK, changes which mask resolves next and later reads that order from the text. Shibboleth instead keeps the order fixed and asks the checkpoint to reproduce its response.

Detectors that use a model. Model access can support richer tests than a key–token hash. Likelihood-ratio detectors compare watermarked and base conditionals [18]; access to the model can improve a Neyman–Pearson score over a key-only Gumbel statistic [12]; GaussMark verifies a keyed weight perturbation with a forward–backward pass [7]; and model-theft watermarks query a suspect system [43]. Outside watermarking, DetectGPT scores text using log probabilities from the model of interest and perturbed versions of the passage; replacing the source model with a surrogate weakens detection [30]. These model-assisted detectors are most natural when a provider or central verifier can afford model access. Shibboleth shares that operational setting but adds a keyed generation channel and a document-level test. It leaves the checkpoint unchanged and keys the contexts presented to it, so one deployed model can serve many keys.

Finite-sample detection and editing. Watermark tests increasingly replace asymptotic z -scores with exact tails, permutation tests, or corrected maxima over windows [12, 22, 24]. After editing, a statistic that concentrates on surviving regions can outperform a document-wide sum [27]. Shibboleth uses an exact binomial tail for the unedited detector and compares document-wide, block-local, and windowed aggregation under editing in Section 4.2.

3 Shibboleth: Watermarking by Model Response

3.1 Masked Decoding

A masked DLM predicts many unresolved positions at once rather than exposing a unique “next token.” Given a sequence containing masks, it returns a distribution at every unresolved position. Let x be the prompt, $y = (y_1, \dots, y_N)$ the continuation, and I the positions already revealed. At any masked position $i \notin I$, the model provides $p_\theta(y_i | y_I, x)$. One forward pass over the full sequence provides these distributions for every unresolved position; we count this as one full-sequence model call.

We study the blockwise schedule used by LLaDA [2, 31]. Blocks finish from left to right, while a confidence rule decides which masks inside the active block commit at each step. We write B_b for the active block and F_b for the completed prefix before it. Shibboleth keeps both the block order and this confidence rule.

3.2 Problem Setup

We target keyed provenance detection: distinguish text generated with a secret document key from text produced independently of that key, at a false-positive rate fixed in advance. The watermark may change which token is sampled, but should not replace the model’s decoder or its schedule.

What the two parties retain. The embedder has the model, prompt, secret key K , and target bits. The verifier later has the completed sequence, the same checkpoint, K , and public parameters—but no generation trace, stored logits, random seed, or record of rejected samples. A nonce derives a fresh document key $K_d = \text{HMAC}(K, \text{nonce}_d)$ [5]. Separate hash domains use K_d for probe construction, carrier assignment, and the requested response directions.

Guarantee and scope. For any text produced independently of K_d , including output from another sampler, a detector run at level α fires with probability at most α . This guarantee is finite-sample and permits the method to choose a data-dependent number of carriers. It does not promise detection after editing, key disclosure, adaptive detector queries [19], or replay with a numerically different checkpoint.

Why write during generation? Editing only after completion offers too little freedom at most positions. Across 36 post-hoc substitution settings, the median selected position has only one candidate within a 3-nat likelihood budget; the best setting below a $1.35 \times$ PPL budget reaches 0.10 TPR at 1% FPR. Shibboleth therefore writes while the model’s live distribution is still available.

The design follows three requirements:

1. **Replayability.** From the completed sequence, the verifier must reconstruct the same masked context that the embedder used.

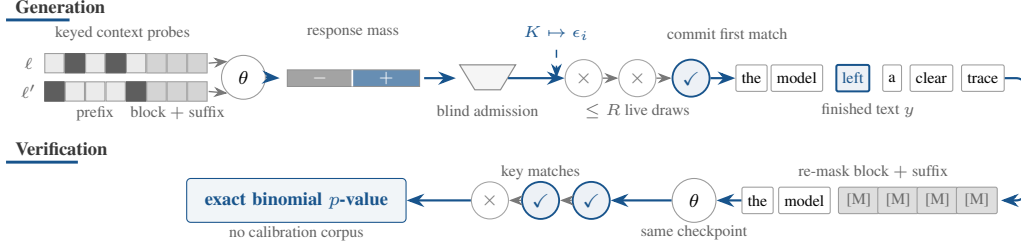


Figure 2: **One block of generation and verification.** Two hidden-prefix views separate candidates by response direction. Carrier selection occurs before the key supplies its orientation; generation commits the first matching draw. Verification rebuilds the same masked state and returns an exact binomial p -value.

- 123 2. **Blind selection.** Carrier positions must be fixed before the key reveals which response direction
124 counts as a match.
- 125 3. **Schedule preservation.** Blocks and within-block commit order must remain those of the refer-
126 ence decoder.

127 Replayability makes a generation-time response measurable later. Blind selection keeps adaptive
128 carrier discovery from biasing the detector, while schedule preservation limits changes to the deployed
129 generator. The next three subsections implement these requirements.

130 3.3 Probing the Model and Selecting Carriers

131 For each unfinished block, Shibboleth asks the model the same question under several hidden-prefix
132 views: how likely is each candidate token when this part of the earlier context is missing? A
133 candidate’s response is the change between two such views. A useful carrier is a position where the
134 model still assigns probability mass to candidates with both response directions (Figures 2 and 3).

135 Formally, Shibboleth constructs L pseudorandom probe patterns $D_{b,\ell} \subset F_b$, where each pattern
136 specifies prefix positions to hide. It also masks the current block B_b and every later position. One
137 model evaluation for pattern ℓ then gives the log-probability of vocabulary item v at position i :

$$\lambda_{i,\ell}(v) = \log p_\theta(v \mid F_b \setminus D_{b,\ell}; B_b \text{ and suffix masked}). \quad (1)$$

138 The block and suffix are hidden because neither was available when the probe was created. The
139 verifier can therefore reconstruct the same state from the completed sequence instead of seeing later
140 tokens that would alter the probabilities. Probing a block costs $L + 1$ full-sequence model calls: L
141 hidden-prefix views and one view with no prefix positions hidden.

142 For a pair (ℓ, ℓ') , define the response $g_i^{\ell,\ell'}(z) = \lambda_{i,\ell'}(z) - \lambda_{i,\ell}(z)$. Let p_i^{base} be the block-masked
143 distribution with no hidden prefix. The probability mass on either response direction, and a symmetric
144 score for the pair, are

$$q_{i,\pm}^{\ell,\ell'} = \sum_z p_i^{\text{base}}(z) \mathbf{1}[\pm g_i^{\ell,\ell'}(z) > 0], \quad S_i^{\ell,\ell'} = 2 \min(q_{i,+}^{\ell,\ell'}, q_{i,-}^{\ell,\ell'}). \quad (2)$$

145 The score is twice the smaller mass: it is near one when the two directions are balanced and near zero
146 when either direction is absent. At each position, Shibboleth keeps the pair with the largest S_i and
147 selects the position when $S_i > s_{\min}$, subject to an optional per-block cap. Choosing the best pair
148 raises the measured mean score from 0.28 for a fixed near/far pair to 0.45.

149 Crucially, swapping ℓ and ℓ' leaves S_i unchanged while reversing every response sign. Carrier
150 selection therefore reveals nothing about which sign will later be rewarded. Figure 3 makes this
151 ordering explicit.

152 After the carrier is fixed, a separate hash domain supplies an orientation $\epsilon_i \in \{-1, +1\}$. A target
153 $w_i \in \{-1, +1\}$ represents the provenance or payload bit. Together they decide which response
154 direction counts as a match:

$$m_i(z) = \mathbf{1}[w_i \epsilon_i g_i(z) > 0]. \quad (3)$$

155 If $g_i(z) = 0$, a second domain-separated bit assigns the outcome. Such a tie cannot be steered, but it
156 remains an independent fair outcome for detection.

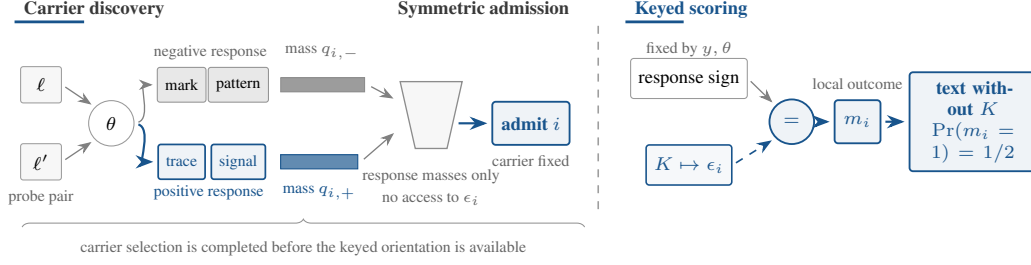


Figure 3: **Selection precedes the keyed orientation.** The probe pair divides candidates by response direction. Position i is selected from the two masses alone; only then does K provide ϵ_i . For text produced independently of K , the two signs agree with probability $1/2$. Token labels and bar widths are schematic.

157 3.4 Embedding Without Changing the Schedule

158 The reference decoder still chooses the next position and supplies its live distribution p_i . Unselected
 159 positions are sampled once as usual. At a selected carrier, Shibboleth accepts a candidate only if it
 160 has the requested response and is not too far below the most likely token:

$$A_i = \left\{ z : w_i \epsilon_i g_i(z) > 0 \text{ and } p_i(z) \geq \kappa \max_v p_i(v) \right\}. \quad (4)$$

161 It draws from the same p_i up to R times and commits the first member of A_i . If every proposal
 162 misses, it takes one fresh unchecked draw. All retries reuse p_i , so they add no model call and do not
 163 alter the commit order.

164 Let $a_i = p_i(A_i)$ be the mass that passes both conditions, and let $q_i = \sum_z p_i(z) m_i(z)$ be the chance
 165 that an unchecked draw scores as a match. The committed token comes from $p_i(\cdot \mid A_i)$ with
 166 probability $1 - (1 - a_i)^R$ and from p_i otherwise. Its match probability is therefore

$$\Pr[m_i = 1] = 1 - (1 - a_i)^R (1 - q_i). \quad (5)$$

167 With no plausibility floor and no ties, $a_i = q_i$, giving $1 - (1 - q_i)^{R+1}$. Equation (5) separates the two
 168 controls: R determines how many chances the sampler has to find an acceptable token, while κ limits
 169 how unlikely that token may be relative to the live argmax. Larger R can strengthen detection but
 170 changes the sampling distribution; κ does not constrain the unchecked fallback or guarantee semantic
 171 quality, which we measure separately.

172 3.5 Watermark Detection

173 The verifier re-masks each completed block and suffix, reconstructs the probe pairs and carriers, and
 174 evaluates $m_i(y_i)$. It needs no rejected samples or generation trace. For provenance detection, the
 175 selected carriers form a pool P with $w_i = +1$; a payload implementation would keep a separate pool
 176 for each message bit. The document statistic and its one-sided p -value are

$$T = \sum_{i \in P} m_i, \quad p_{\text{det}} = \Pr[\text{Binomial}(|P|, \frac{1}{2}) \geq T]. \quad (6)$$

177 Figure 4 follows a five-carrier example from replayed outcomes to the exact upper-tail probability.

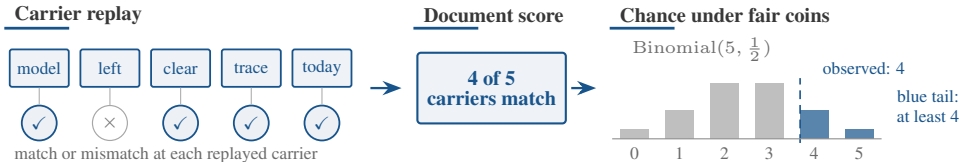


Figure 4: **Document-level detection.** The verifier replays each carrier and records a match. Here four of five match; the blue bars give the exact chance of observing at least four matches under fair coins.

178 **Proposition 1** (Exact false-positive control). Assume the hash domains behave as independent
 179 random functions. For any text written without the document key, condition on the selected carriers,

probe pairs, responses, and target bits. The orientation bits remain independent fair coins. Hence the m_i are independent Bernoulli($\frac{1}{2}$) variables, $T \sim \text{Binomial}(|P|, \frac{1}{2})$, and thresholding Eq. (6) at α gives false-positive rate at most α for every document length and carrier count.

Why adaptive carrier selection is allowed. The text and model may determine the number of carriers, the probe pair, and the response size. What matters is that each choice is fixed before the independent orientation bit is used. Flipping ϵ_i then flips the match whenever $g_i(y_i) \neq 0$; the tie rule supplies a fresh fair bit otherwise. The detector therefore needs neither an asymptotic approximation nor a calibration corpus. The guarantee still assumes text produced independently of the key and domain separation among the key’s several roles.

4 Experiments

The evaluation asks four questions: how strongly the watermark can be detected, whether one setting transfers across checkpoints, what it costs in computation and output quality, and how editing changes detection power.

Models and data. We evaluate LLaDA-8B-Instruct [31] and Dream-7B-Instruct [42] in FP16 on one TITAN RTX. Detectability uses C4 *realnewslike* prompts [36], truncated to 300 characters, and 512-token continuations divided into 32-token blocks. We correct Dream’s one-position logit offset in its wrapper, then use the same position convention for generation and replay.

Decoders and settings. Shibboleth and dgMARK use each model’s confidence schedule with multinomial sampling at temperature 0.8; dgMARK additionally uses top- $k = 3$. KGW requires strictly left-to-right decoding. Shibboleth uses $L = 8$ probe patterns, threshold $s_{\min} = 0.5$, retry budget R , and optional plausibility floor κ . We evaluate provenance ($w_i = +1$), not payload recovery. Here one model call means one forward pass over the current full sequence. With 16 blocks and $L + 1 = 9$ probe views, verification replays $16 \times 9 = 144$ calls; generation uses 512 ordinary decoding calls plus the same 144 probe calls, for 656 in total.

Metrics. Detection is TPR at the analytic FPR thresholds in Eq. (6). Perplexity is scored by GPT-2-large [35] as $\exp(\text{mean}(\text{NLL}_{\text{wm}} - \text{NLL}_{\text{unwm}}))$. A PPL ratio therefore compares a method with its own unwatermarked decoder; it is not a semantic-quality score and is only indicative across different decoders. The baselines are KGW [21] and dgMARK [17].

4.1 Main Results and Analysis

Retries trade sampling distortion for detection. Table 1 gives the 512-token detection, PPL, and model-call comparison. On LLaDA, $R = 1$ reaches 0.43 TPR at 1% FPR. On 30 held-out prompts, $R = 2, 4, 8$ increase TPR to 0.73, 0.83, 0.87, consistent with Eq. (5). Retries add no model evaluation, but without a floor the PPL ratio rises from $1.14\times$ at $R = 2$ to $1.61\times$ at $R = 8$. The floor removes low-probability guided draws: $(R, \kappa) = (8, 0.1)$ gives 0.70 TPR at $0.87\times$ PPL, and $(16, 0.05)$ gives 0.80 at $0.74\times$. Ratios below one mean only that GPT-2-large prefers the retained high-probability tokens.

Transfer is checkpoint-dependent. We transfer all LLaDA settings to Dream without retuning. KGW falls to 0.57 TPR at 1% FPR and produces 84% duplicate bigrams under forced left-to-right decoding. Direct and three-beam dgMARK reach 0.37 and 0.30 TPR. Shibboleth at $(16, 0.05)$ reaches 0.90 at both reported FPR levels. Its PPL ratio is $0.79\times$ at $(8, 0.1)$ and $1.09\times$ at $(16, 0.05)$, based on 8–13 valid pairs per arm. Detection transfers here; the PPL-minimising setting does not.

Task accuracy exposes decoder-specific costs. Table 2 shows each method’s own unwatermarked and watermarked scores side by side; Appendix Table 5 gives the full grid. This pairing matters because each method uses a different decoding procedure. For Shibboleth, changes are at most 0.06 across the measured MMLU, GSM8K, and HumanEval cells. For dgMARK, HumanEval on LLaDA changes from 0.409 to 0.341 direct and to 0.280 with three-beam lookahead; on Dream, MMLU changes from 0.740 to 0.570 and GSM8K from 0.840 to 0.560. These task outputs are also shorter: on LLaDA GSM8K, Shibboleth and direct dgMARK reach 0.12 TPR at 1% FPR, three-beam

Method	Setting	PPL ratio ↓	TPR at FPR ↑		Full-sequence model calls ↓	
			1%	0.1%	Generation	Verification
<i>LLaDA-8B-Instruct</i>						
KGW [21]	$\delta = 1$	1.22×	0.90	0.83	512	0
dgMARK [17]	$k = 1$	1.45×	0.83	0.80	512	0
dgMARK [17]	+3-beam	1.21×	0.93	0.93	~2048	0
Shibboleth	$R = 1$	0.97×	0.43	0.20	656	144
Shibboleth	$R = 8, \kappa = 0.1$	0.87×	0.70	0.60	656	144
Shibboleth	$R = 16, \kappa = 0.05$	0.74 ×	0.80	0.60	656	144
<i>Dream-7B-Instruct</i>						
KGW [21]	$\delta = 1$	1.84×	0.57	0.37	512	0
dgMARK [17]	$k = 1$	1.75×	0.37	0.27	512	0
dgMARK [17]	+3-beam	0.71 ×	0.30	0.30	~2048	0
Shibboleth	$R = 1$	1.34×	0.40	0.27	656	144
Shibboleth	$R = 8, \kappa = 0.1$	0.79×	0.70	0.67	656	144
Shibboleth	$R = 16, \kappa = 0.05$	1.09×	0.90	0.90	656	144

Table 1: **Detectability and cost at 512 tokens.** PPL ratio is measured against each method’s own unwatermarked run and should be interpreted within that method. Shibboleth uses $n = 30$ documents ($n = 20$ for $R = 16$); Dream PPL uses 8–13 successfully scored document pairs per setting. Settings chosen on LLaDA are transferred to Dream without retuning. One model call is one forward pass over the current full sequence; these counts measure algorithmic work, not runtime. δ is KGW’s logit bias and k is dgMARK’s candidate count.

Method	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑
<i>LLaDA-8B-Instruct</i>			
KGW [21]	0.530→0.535	0.640→0.740	0.329→0.293
dgMARK [17]	0.605→0.530	0.740→0.660	0.409→0.341
dgMARK +3-beam	0.605→0.530	0.740→0.660	0.409→0.280
Shibboleth	0.595→0.575	0.640→0.700	0.335→0.384
<i>Dream-7B-Instruct</i>			
KGW	0.640→0.625	0.700→0.700	—
dgMARK	0.740→0.570	0.840→0.560	—
dgMARK +3-beam	0.740→0.645	0.840→0.560	—
Shibboleth	0.695→0.710	0.660→0.680	—

Table 2: **Task accuracy before and after watermarking.** Each cell is unwatermarked → watermarked under that method’s decoder. MMLU uses 200 questions, GSM8K 50 problems, and HumanEval 164 executed tasks. Approximate standard errors are 0.035, 0.07, and 0.037, respectively; — denotes an unmeasured pair. Shibboleth uses $R = 16, \kappa = 0.05$.

228 dgMARK 0.44, and KGW zero. The 512-token detection rates therefore do not describe 256-token
229 answers.

230 **Checkpoint binding costs model calls.** On LLaDA, Shibboleth’s 0.80/0.60 TPR at 1%/0.1%
231 FPR trails three-beam dgMARK’s 0.93/0.93 and KGW’s 0.90/0.83. The token-local methods
232 need no model evaluation during detection; Shibboleth needs 144. KGW’s left-to-right decoder
233 also repeats heavily: before bigram deduplication, 37% of its unwatermarked documents cross the
234 decision threshold; after deduplication, two of thirty still cross the 0.1% threshold. With cohorts of
235 20–30 documents, TPR differences near 0.1 remain imprecise.

236 4.2 Calibration and Robustness

237 **Observed false-positive rates are below nominal levels.** Appendix Table 3 reports the per-key
238 results. Across 20 keys, with 120 unwatermarked LLaDA generations per key, mean FPR is 0.0121,
239 0.0037, and 0.0000 at nominal 0.10, 0.05, and 0.01. The largest per-key rates are 0.0583, 0.0250,
240 and 0.0000. Every Dream unwatermarked cohort has zero false positives at 1%.

241 **Dispersed edits break prefix synchronisation.** Appendix Table 4 summarises the editing results.
242 At $R = 1$, either same-model argmax re-denoising or deleting a random 10% of positions reduces

243 TPR at 1% FPR from 0.35 to 0.05 on 20 documents. At $R = 16$, $\kappa = 0.05$ and 1024 tokens
244 ($n = 16$), a random 5% edit leaves TPR at 0.88, while a random 10% edit reduces it to 0.50; false-
245 positive control remains valid on identically edited unwatermarked text. In contrast, re-denoising one
246 contiguous span covering 10% or 30% of positions leaves TPR at 0.88, because carriers before the
247 span are unchanged.

248 Neither tested aggregation change improves the random-edit result. A Bonferroni-combined per-block
249 test gives 0.38 TPR versus 0.50 for document-wide counting under random 10% editing, and 0.69
250 versus 0.88 without editing. A 128-token conditioning window gives 0.44 versus 0.50 under random
251 10% editing. Copy-paste-style sparse survival was not evaluated.

252 5 Discussion

253 **Deployment role.** Shibboleth is designed for model providers, centralised verification services, and
254 auditors of high-value documents: settings in which the production decoder should remain unchanged
255 and a verifier can retain the checkpoint. It is not a replacement for inexpensive token-local detection.
256 A practical system could use a token-local test for routine traffic and reserve checkpoint replay for
257 cases where model-specific provenance matters. This binding is both the method’s distinction and its
258 cost. The checkpoint identity, numerical precision, and replay implementation must all be retained;
259 our experiments do not establish transfer across numerically different copies.

260 **Interpreting a detection result.** A low p -value has a narrow interpretation: under the registered
261 key and replay procedure, the document contains more checkpoint-consistent carrier responses than
262 expected by chance. It does not identify the person who requested the text, establish that every
263 passage came from the model, or rule out later editing. Those questions require account records,
264 cryptographic signatures, or separate forensic analysis. A verifier should report the p -value and
265 carrier count, along with the key identifier, checkpoint hash, numerical precision, probe construction,
266 and decision threshold. The carrier count helps distinguish an inconclusive short document from a
267 response that disagrees with the checkpoint.

268 **Editing and quality.** Dispersed edits alter many later prefixes and therefore many reconstructed
269 probes; a contiguous span leaves all earlier carriers intact. This asymmetry points to content-anchored
270 probes, rather than absolute prefix positions, as a promising route to editing tolerance. The block-
271 local and windowed variants tested here did not improve detection after random edits. Quality
272 measurements also require restraint: a PPL ratio below one does not imply better text, and the task
273 cohorts are too small to support claims about modest score changes. Under the measured settings,
274 Shibboleth’s task-score changes are small relative to the decrease observed for direct dgMARK on
275 Dream, but its verifier requires substantially more computation.

276 6 Conclusion

277 Shibboleth establishes model response as a watermark channel for masked DLMs. Its signal is a
278 reproducible change in model probability under hidden context, rather than membership in a keyed
279 token subset. The central lesson is the ordering: select a carrier with a symmetric response score,
280 then reveal the keyed orientation. This separation yields an exact binomial test even when the text
281 and model determine how many carriers are available.

282 The method complements token-local watermarking rather than replacing it. It preserves the deployed
283 decoding schedule and reaches 0.80 TPR on LLaDA and 0.90 on Dream with settings transferred
284 without retuning. In return, the verifier must replay the same checkpoint 144 times per 512-token
285 document. Token-local detectors remain cheaper and reach higher TPR on LLaDA.

286 **Limitations.** The false-positive guarantee does not imply detection after editing. Random 10%
287 re-denoising reduces TPR from 0.35 to 0.05 at $R = 1$ and from 0.88 to 0.50 at $R = 16$, although
288 one contiguous edited span leaves the measured TPR unchanged. Replay also requires the same
289 checkpoint and numerical precision. Evaluation covers two masked DLMs, one prompt corpus,
290 cohorts of 16–50 documents, and provenance only; payload recovery is not measured.

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402 A Additional Tables

403 Table 3 reports the measured per-key false-positive rate of the deployed decision rule. Table 4
 404 summarises the post-editing evaluation under the deployed pooled detector. Table 5 expands Table 2
 405 into the full task-accuracy grid, including the greedy-decoding cells.

Nominal α	Mean FPR	Highest per-key FPR
0.10	0.0121	0.0583
0.05	0.0037	0.0250
0.01	0.0000	0.0000

Table 3: **Observed false-positive rates across keys.** The evaluation uses 20 keys and 120 unwatermarked generations per key under the deployed decision rule.

		TPR at FPR $\uparrow$	
Attack	Strength	1%	0.1%
<i>Configuration A: $R = 1$, 512 tokens ($n = 20$)</i>			
None	—	0.35	—
Re-denoising (same model, argmax)	10% of positions	0.05	—
Deletion	10% of positions	0.05	—
Insertion	10% of positions	—	—
Substitution (random)	10% of positions	—	—
Copy-paste	25% kept, 1 segment	—	—
Paraphrase (DIPPER) [23]	lex 20, order 0	—	—
<i>Configuration B: $R = 16$, $\kappa = 0.05$, 1024 tokens ($n = 16$)</i>			
None	—	0.88	0.88
Re-denoising (random positions)	5% of positions	0.88	0.75
	10% of positions	0.50	0.38
Re-denoising (contiguous span)	10% of positions	0.88	0.88
	30% of positions	0.88	0.88

Table 4: **Detection after editing.** The pooled detector uses the same analytic thresholds as Table 1. Random edits disrupt later prefixes; a contiguous span preserves all earlier carriers. Because the cohorts are small, one document changes TPR by 0.05 when $n = 20$ and about 0.06 when $n = 16$. An em dash denotes an unmeasured setting.

Model	Method	Greedy decoding			Multinomial decoding		
		MMLU Acc $\uparrow$	GSM8K Acc $\uparrow$	HumanEval Pass@1 $\uparrow$	MMLU Acc $\uparrow$	GSM8K Acc $\uparrow$	HumanEval Pass@1 $\uparrow$
LLaDA-8B-Instruct	Non-watermarked	0.605	0.740	0.409	0.595	0.640	0.335
	KGW [21]	—	—	—	0.535	0.740	0.293
	dgMARK [17]	—	—	—	0.530	0.660	0.341
	dgMARK +3-beam	—	—	—	0.530	0.660	0.280
	Shibboleth	—	—	—	0.575	0.700	0.384
Dream-7B-Instruct	Non-watermarked	0.740	0.840	—	0.695	0.660	—
	KGW	—	—	—	0.625	0.700	—
	dgMARK	—	—	—	0.570	0.560	—
	dgMARK +3-beam	—	—	—	0.645	0.560	—
	Shibboleth	—	—	—	0.710	0.680	—

Table 5: **Text generation quality, full grid.** Accuracy on MMLU [15] and GSM8K [11] and pass@1 on HumanEval [8] under greedy and multinomial decoding, with the unwatermarked run of each model as the reference row of its block. Shibboleth uses $R = 16$, $\kappa = 0.05$; 50 problems per GSM8K cell, so differences in individual scores should be interpreted cautiously. Each watermark decodes under its own regime—KGW left-to-right (its green list requires a resolved prefix; its own control scores 0.640), dgMARK under confidence decoding (its argmax control is the greedy reference cell)—so rows are read against their own controls, not across. An em dash denotes an unmeasured setting.

406 B Where the Watermark Can Be Written

407 Figure 5 replays the verifier’s response bank over saved unwatermarked 512-token outputs and maps
 408 the admission score of Eq. (2) across every position. On this checkpoint the channel is wide: 97% of
 409 positions retain probability mass on both response signs and pass the $s_{\min} = 0.5$ gate, and the sharp,
 410 unwritable positions concentrate at the first tokens of a block, where the block-masked conditional is
 411 most constrained. The two showcased conditionals make the criterion concrete: a good carrier is a
 412 position where several plausible tokens disagree about the response sign, so committing either sign
 413 stays on-distribution; a one-sided position has a single dominant continuation, and the gate rejects it
 414 before the keyed orientation is revealed.

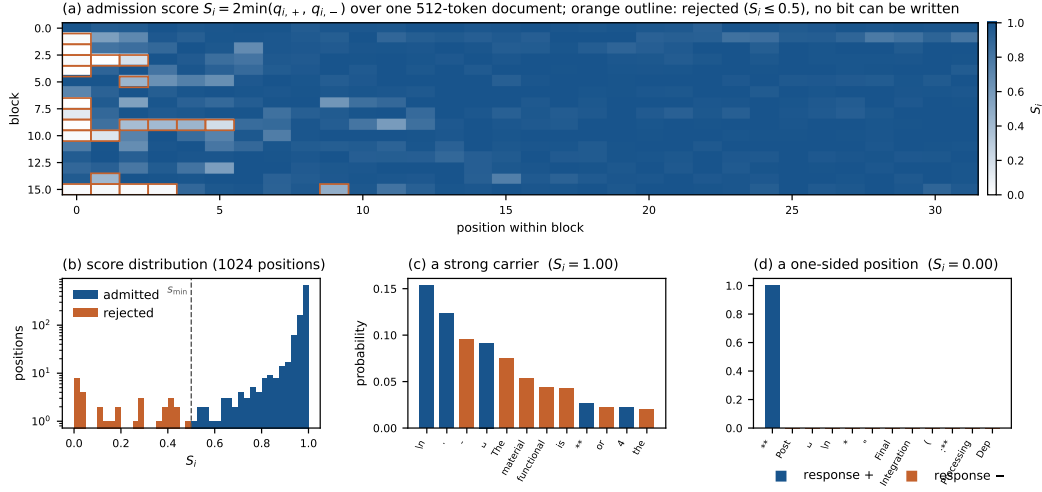


Figure 5: **Where the watermark can be written.** The verifier’s response bank replayed over an unwatermarked 512-token LLaDA output. (a) Admission score $S_i = 2 \min(q_{i,+}, q_{i,-})$ at every position of one document; orange outlines mark the rejected positions ($S_i \leq s_{\min} = 0.5$), where the block-masked conditional concentrates on one response sign and no bit can be written. (b) Across 1024 positions of two documents, 97% pass the gate; since $q_{i,+} + q_{i,-} \approx 1$ on real text, $S_i \approx 1 - |q_{i,+} - q_{i,-}|$, and the gate removes the sharp tail. Thirty of the thirty-one rejected positions sit within the first six tokens of a block, where the conditional is most constrained. (c) A strong carrier: the conditional spreads its mass over tokens of both response signs, so either sign can be committed from a plausible token. (d) A one-sided position: a single dominant token fixes the response, and the position is rejected before the key’s orientation is consulted.